

Localization and Object Detection

A fixed dense tensor becomes a scored set of boxes

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ES 667 · Deep Learning · IIT Gandhinagar

**L8B kept spatial features;
detection must spend them**

L8B answered “what”; today the head must preserve “where”

L8B · CLASSIFICATION

backbone $\rightarrow 7 \times 7 \times 512 \rightarrow$ global
average $\rightarrow$ one label

$\rightarrow$

L10 · DETECTION

spatial features $\rightarrow$ many class
scores + many boxes $\rightarrow$ a variable
set

L8B answered “what”; today the head must preserve “where”

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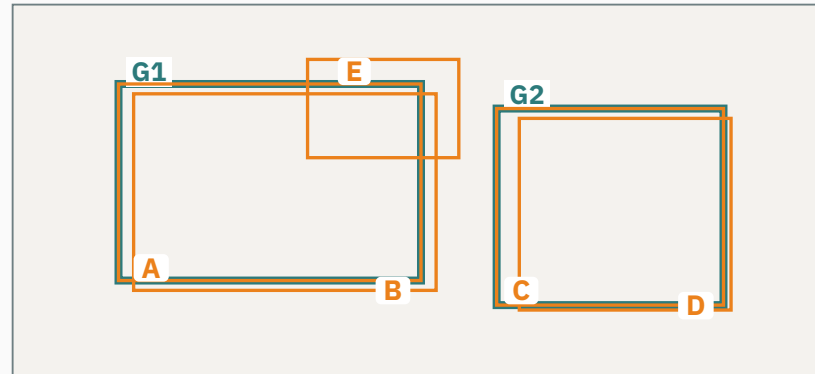
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Global averaging buys image-level invariance by discarding location. Detection reuses the backbone, but changes the head, targets, geometry, and evaluation.

Commit: which operations turn five candidates into trustworthy detections?

QUESTION

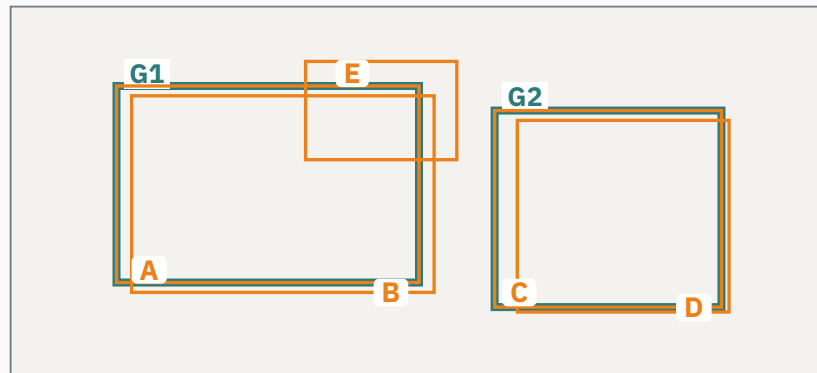
HELD CASE · OPENING SCENE A dense head emits A .95, B .90, E .89, C .88, D .75 for two cup truths G_1, G_2 .



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A · THRESHOLD

keep every high score;
duplicates disappear
automatically

B · GEOMETRY

decode → score → NMS;
then match for evaluation

C · LOSS

run the training loss again
at inference

The 80-minute route has one cumulative spine

By the end, you should be able to turn a dense output tensor into trained candidates, filtered detections, and a defensible AP number.

CORE · 55 MIN

box contract → IoU →
dense head →
assignment + loss → NMS
→ matching + AP →
revisit

SHOULD · 15 MIN

Smooth L_1 /GIoU → two-
stage vs one-stage →
FPN → COCO protocol

OPTIONAL · 10 MIN

anchor-free heads →
Soft-NMS / set prediction
→ primary provenance

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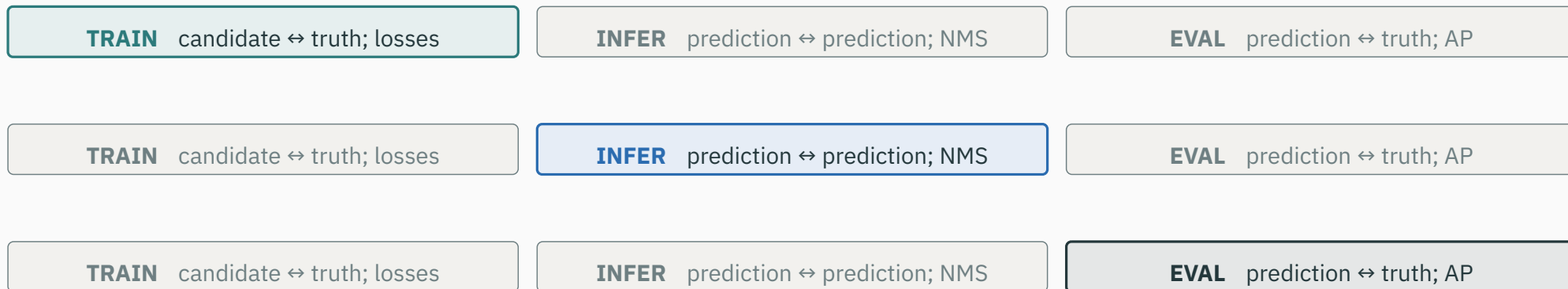
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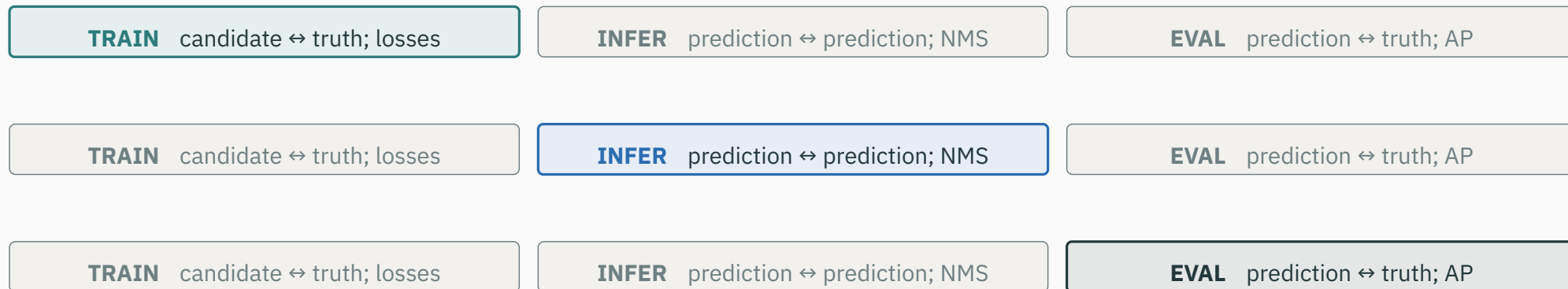
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— given / eval — truth / train — raw candidate — infer / control — survivor / TP — suppressed / FP

Three clocks reuse boxes—but never the same comparison



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TRAIN

assign candidate ↔ truth

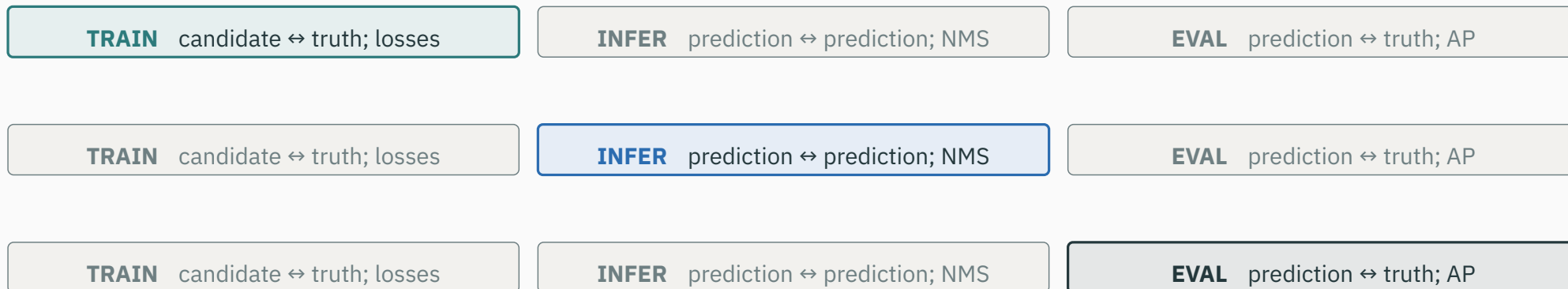
INFER

compare prediction ↔ prediction

EVAL

match prediction ↔ truth

Three clocks reuse boxes—but never the same comparison



TRAIN

assign candidate ↔ truth

INFER

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EVAL

match prediction ↔ truth

The clock strip will remain visible whenever “IoU threshold” could otherwise mean three different things.

**A box is four numbers with an
explicit convention**

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classifier: $x \rightarrow \hat{p} \in \Delta^{K-1}$

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CLASS

$\hat{y}_i$: what is here?

BOX

$\hat{b}_i \in \mathbb{R}^4$: where is it?

SCORE

s_i : how confident?

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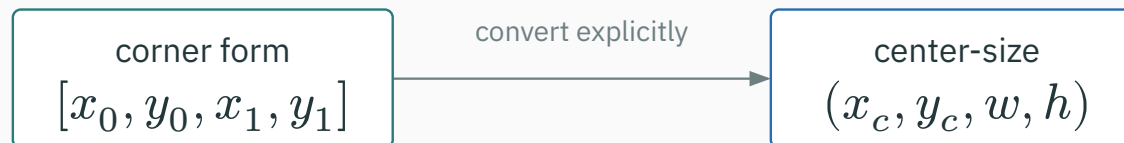
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A fixed network tensor may contain many candidate triples; thresholding and NMS make the returned set variable-length.

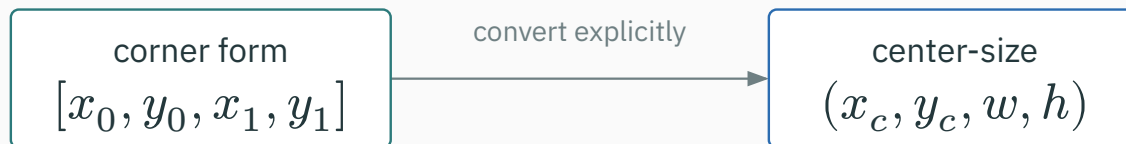
Our coordinate convention prevents the silent “+1” bug

HELD CASE · BOX CONVENTION Every box is $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ with width $x_{\max} - x_{\min}$ and height $y_{\max} - y_{\min}$.



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We use continuous boundaries—or equivalently half-open pixel boxes. Area is $(x_1 - x_0)(y_1 - y_0)$: there is no +1.

The held truths fix every later area calculation

HELD CASE · GROUND TRUTH $G_1 = [10, 10, 50, 50]$ and $G_2 = [60, 15, 90, 55]$ on a 100×80 canvas.

G_1

$$w = 50 - 10 = 40$$

$$h = 50 - 10 = 40$$

$$|G_1| = 1,600$$

G_2

$$w = 90 - 60 = 30$$

$$h = 55 - 15 = 40$$

$$|G_2| = 1,200$$

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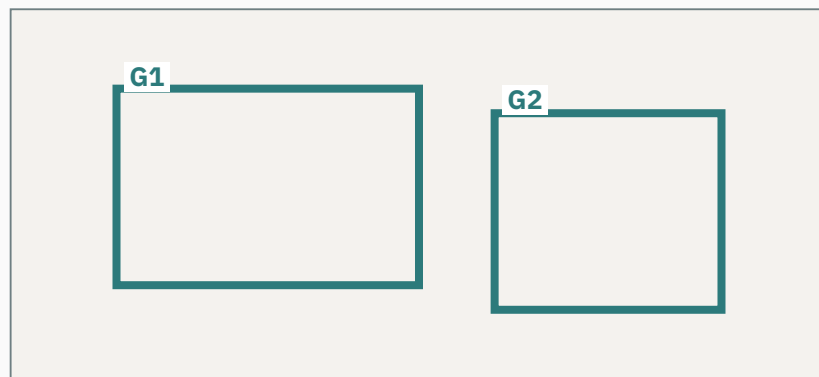
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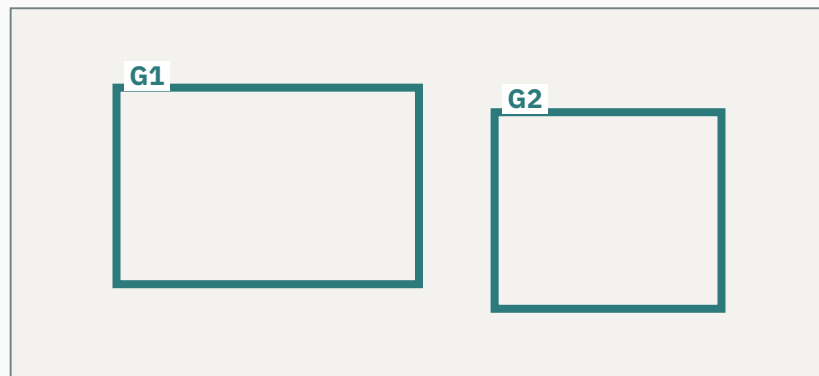
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These exact rectangles anchor every later IoU, suppression decision, and match.

A localization head shares evidence, then asks two questions

SHARED LOCAL FEATURE

$$h = f_{\theta(x)}$$

→

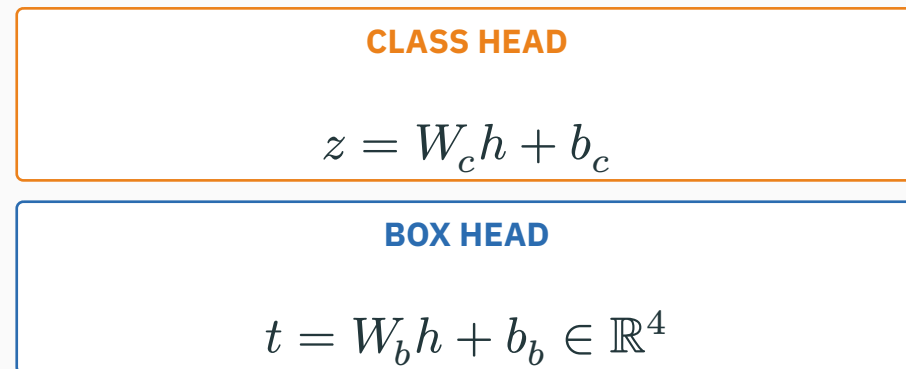
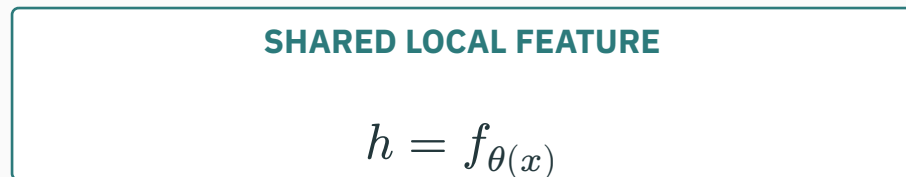
CLASS HEAD

$$z = W_c h + b_c$$

BOX HEAD

$$t = W_b h + b_b \in \mathbb{R}^4$$

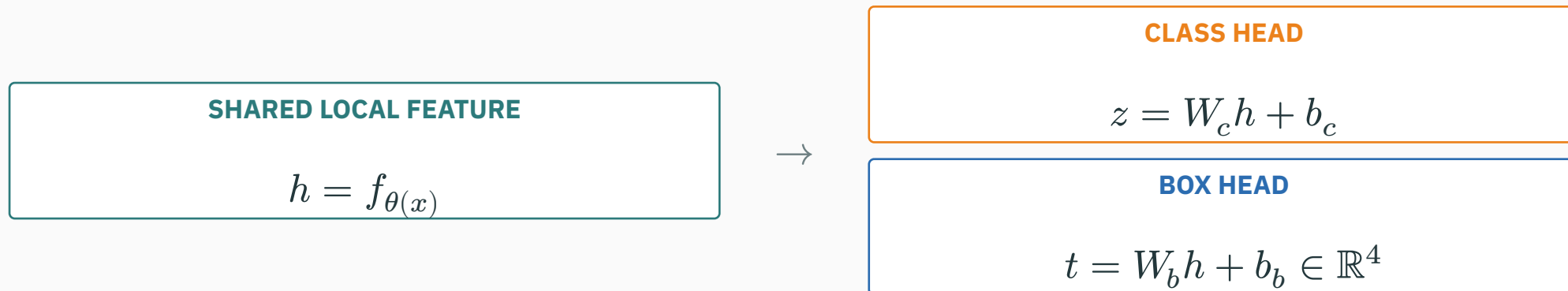
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what is here?

how should the reference move?

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Detection is multi-task learning repeated at spatial locations—not a classifier with four unexplained extra outputs.

Commit: what offsets move this anchor to its matched truth?

QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

HELD CASE · POSITIVE ANCHOR Reference $a = (50, 50, 40, 40)$; matched target $b = (58, 46, 80, 40)$, both in center-size form.

TRANSLATE

$$t_x = \frac{x - x_a}{w_a}$$
$$t_y = \frac{y - y_a}{h_a}$$

RESIZE

$$t_w = \log\left(\frac{w}{w_a}\right)$$
$$t_h = \log\left(\frac{h}{h_a}\right)$$

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Calculate all four before the answer. Which offset must be zero? Which must be $\log 2$?

The anchor target is a scaled correction, not an absolute box

A ANSWER

TRAIN candidate $\leftrightarrow$ truth; losses

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$$t_x = \frac{58-50}{40} = 0.20$$
$$t_y = \frac{46-50}{40} = -0.10$$

$$t_w = \log\left(\frac{80}{40}\right) = \log 2 \approx 0.693$$
$$t_h = \log\left(\frac{40}{40}\right) = 0$$

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$$t = (0.20, -0.10, 0.693, 0)$$

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Translations are measured in anchor sizes; log-scale changes keep decoded widths positive and make zero mean “unchanged.”

Decoding must recover the same geometry

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

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$$\hat{x} = x_a + t_x w_a = 50 + 0.20(40) = 58$$

$$\hat{y} = y_a + t_y h_a = 50 - 0.10(40) = 46$$

$$\hat{w} = w_a \exp(t_w) = 40 \exp(0.693) \approx 80$$

$$\hat{h} = h_a \exp(t_h) = 40 \exp(0) = 40$$

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Encode and decode are inverse contracts; a unit test should recover b before any model is trained.

A YOLO-style anchor head emits nine scalars per candidate

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

HELD CASE · HEAD CONVENTION For this teaching ledger only: one objectness logit + $K = 4$ class logits + four box offsets.

OBJECTNESS

one logit
is anything here?

CLASSES

$K = 4$ logits
which class?

OFFSETS

four numbers
how should the anchor
move?

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Teaching convention only: other heads may fold background into classes or predict box sides.

The dense tensor fixes capacity before the object count is known

D DERIVATION

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HELD CASE · TENSOR LEDGER 20×20 locations, $A = 3$ anchors per location, $K = 4$ classes, nine outputs per anchor.

quantity	calculation	count
candidate boxes	$20 \cdot 20 \cdot 3$	1,200
output scalars	$20 \cdot 20 \cdot 3 \cdot 9$	10,800

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The head always emits 10,800 scalars. Labels and post-processing decide which candidate records matter.

Toy assignment says which candidates receive which losses

TRAIN candidate $\leftrightarrow$ truth; losses**INFER** prediction $\leftrightarrow$ prediction; NMS**EVAL** prediction $\leftrightarrow$ truth; AP

For one truth, three anchors have IoUs 0.76, 0.48, and 0.12. Use the declared **toy** policy $\tau_+ = 0.70$, $\tau_- = 0.30$.

IoU	assignment	loss terms
0.76	positive	objectness + class + box
0.48	ignore	no gradient in this toy policy
0.12	background	objectness only

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Assignment is a training rule, not AP matching and not NMS. Real detectors use architecture-specific thresholds or dynamic matchers.

One positive candidate produces a fully auditable loss

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

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Target $t = (.20, -.10, .693, 0)$; prediction $\hat{t} = (.25, -.05, .60, .05)$. Let $p_{\text{obj}} = .90$, $p_{\text{cup}} = .80$ and $\lambda = 2$.

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$$-\log .90 \approx .105$$

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OBJECTNESS

CLASS

$$-\log .90 \approx .105$$

$$-\log .80 \approx .223$$

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BOX L_1

$$|.05| + |.05| + |-.093| + |.05| = .243$$

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$$\mathcal{L} = .105 + .223 + 2(.243) \approx .815$$

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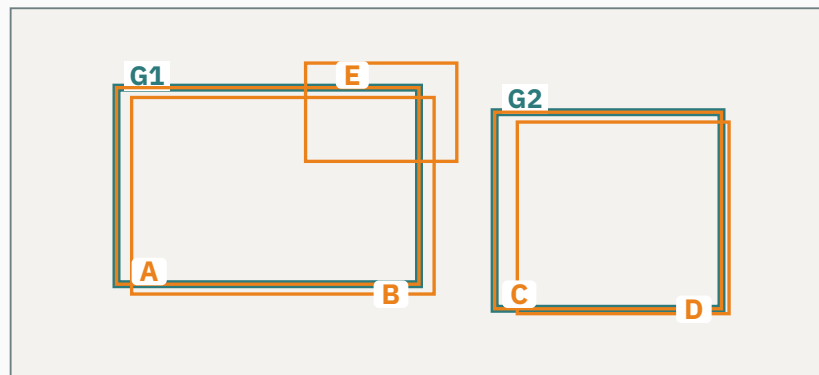
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Write the target, parameterization, reduction, and λ . “Box loss” without those contracts is not reproducible.

**IoU makes box quality
geometric**

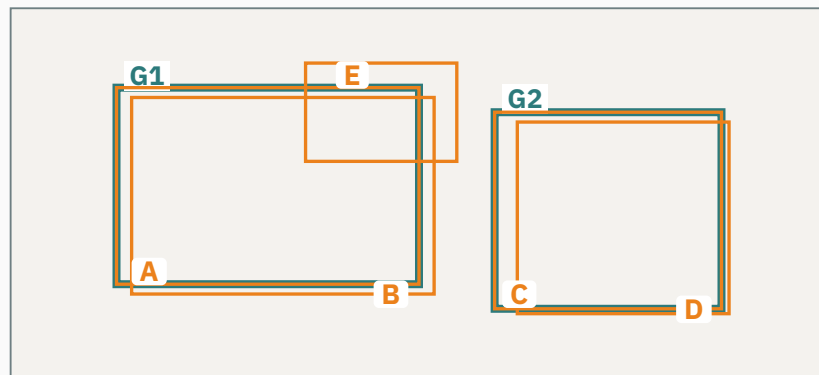
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HELD CASE · HELD PAIR $A = G_1 = [10, 10, 50, 50]$ and $B = [12, 12, 52, 52]$.



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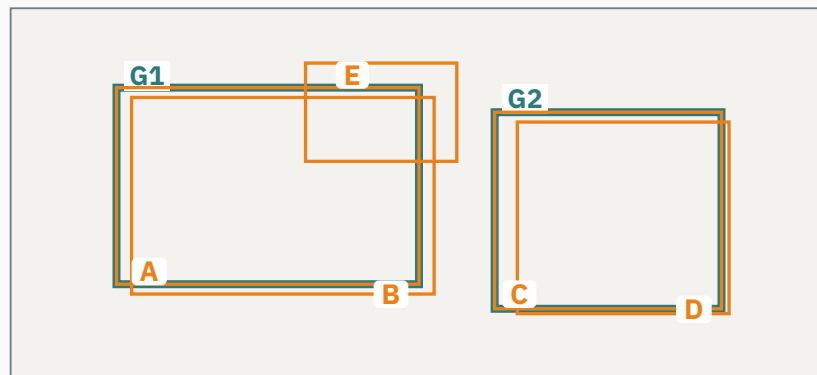
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0: disjoint 1: identical invariant to scaling the whole scene

The held overlap is 0.822, not a visual guess

D DERIVATION

HELD CASE · SAME PAIR $A = [10, 10, 50, 50]$, $B = [12, 12, 52, 52]$; both have area 1,600.

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HELD CASE · SAME PAIR $A = [10, 10, 50, 50]$, $B = [12, 12, 52, 52]$; both have area 1,600.

$$w_I = \min(50, 52) - \max(10, 12) = 38$$

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$$A_I = 38 \cdot 38 = 1,444$$

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$$\text{IoU}(A, B) = \frac{1444}{1756} \approx 0.822$$

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$$\text{IoU}(A, B) = \frac{1444}{1756} \approx 0.822$$

The same 0.822 will later justify suppressing B around kept box A.

Safe IoU clamps disjoint intersections to zero

D DERIVATION

For each axis, $w_I = \max(0, \min(x_1^A, x_1^B) - \max(x_0^A, x_0^B))$ $h_I = \max(0, \min(y_1^A, y_1^B) - \max(y_0^A, y_0^B))$

Then, $A_I = w_I h_I$ $A_U = A_A + A_B - A_I$
 $\text{IoU} = \frac{A_I}{A_U}$

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D DERIVATION

For each axis, $w_I = \max(0, \min(x_1^A, x_1^B) - \max(x_0^A, x_0^B))$ $h_I = \max(0, \min(y_1^A, y_1^B) - \max(y_0^A, y_0^B))$

Then, $A_I = w_I h_I$ $A_U = A_A + A_B - A_I$
 $\text{IoU} = \frac{A_I}{A_U}$

Without $\max(0, \cdot)$, two disjoint boxes can produce two negative side lengths—and a meaningless positive “intersection.”

Safe IoU clamps disjoint intersections to zero

D DERIVATION

For each axis, $w_I = \max(0, \min(x_1^A, x_1^B) - \max(x_0^A, x_0^B))$ $h_I = \max(0, \min(y_1^A, y_1^B) - \max(y_0^A, y_0^B))$

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 $\text{IoU} = \frac{A_I}{A_U}$

Without $\max(0, \cdot)$, two disjoint boxes can produce two negative side lengths—and a meaningless positive “intersection.”

Also assert valid boxes: $x_1 \geq x_0$, $y_1 \geq y_0$, and define the zero-union case explicitly.

SHOULD · equal coordinate error can mean unequal overlap

Two predictions move one boundary by 10 pixels.

Large object: 200×200

a 10-pixel error is 5% of its width.

Small object: 20×20

the same error is 50% of its width.

SHOULD · equal coordinate error can mean unequal overlap

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Coordinate losses see four residuals. IoU sees the region those residuals create.

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Small object: 20×20

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Coordinate losses see four residuals. IoU sees the region those residuals create.

Use coordinate losses for a stable regression signal; report overlap-aware metrics for the geometry the task actually values.

SMOOTH $L_1 \cdot \beta = 1$

quadratic near 0; linear for large residuals

$0.2 \rightarrow 0.02$; $3 \rightarrow 2.5$

GIoU

$$\text{GIoU} = \text{IoU} - \frac{|C|}{|A \cup B|}$$

C : smallest enclosing box

SHOULD · Smooth L_1 and GIoU repair different failures

D DERIVATION

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quadratic near 0; linear for large
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$$\text{GIoU} = \text{IoU} - \frac{\frac{|C|}{(A \cup B)}}{|C|}$$

C : smallest enclosing box

Plain 1 – IoU is flat while boxes are disjoint. GIoU changes with the enclosing gap;
Smooth L_1 instead controls coordinate-residual robustness.

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Plain 1 – IoU is flat while boxes are disjoint. GIoU changes with the enclosing gap; Smooth L_1 instead controls coordinate-residual robustness.

Rezatofighi et al. · Generalized IoU

**TRAIN: dense candidates need
selective supervision**

Dense heads create a foreground–background imbalance

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



Dense heads create a foreground–background imbalance

TRAIN candidate $\leftrightarrow$ truth; losses

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If thousands of easy background candidates contribute ordinary cross-entropy, their count can drown out the rare positives and hard negatives.

Focal loss turns confidence into a gradient-volume control

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

$$\text{FL}(p_t) = -\alpha_{t(1-p_t)}^{\gamma} \log p_t$$

Focal loss turns confidence into a gradient-volume control

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p_t

probability assigned to
the true binary label

γ

how strongly easy
examples are down-
weighted

α_t

optional class-balance
weight

Focal loss turns confidence into a gradient-volume control

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probability assigned to
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how strongly easy
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optional class-balance
weight

| The next numeric sets $\alpha_t = 1$ and $\gamma = 2$ to isolate the focusing factor.

With $\gamma = 2$, the hard example is about $978 \times$ louder

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

Easy: $p_t = .9$

$$\text{CE} = -\log .9 \approx .10536$$

$$(1 - p_t)^2 = .01$$

$$\text{FL} \approx .0010536$$

Hard: $p_t = .2$

$$\text{CE} = -\log .2 \approx 1.60944$$

$$(1 - p_t)^2 = .64$$

$$\text{FL} \approx 1.03004$$

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$$\frac{1.03004}{.0010536} \approx 977.6 \approx 978 \times$$

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$$(1 - p_t)^2 = .64$$

$$\text{FL} \approx 1.03004$$

$$\frac{1.03004}{.0010536} \approx 977.6 \approx 978 \times$$

Focal loss does not delete easy examples; it makes already-correct examples quiet enough that rare mistakes remain audible.

Diagnose training failures as symptom → suspect → test

TRAIN candidate ↔ truth; losses**INFER** prediction ↔ prediction; NMS**EVAL** prediction ↔ truth; AP

symptom	suspect	smallest discriminating test
box loss falls; IoU stays near 0	box format / decode mismatch	round-trip one known anchor and draw decoded boxes
almost no positive anchors	assignment geometry too strict	histogram best IoU per truth; inspect τ_+
objectness predicts background	easy negatives dominate	log pos/neg loss mass; compare focal or sampling

Diagnose training failures as symptom → suspect → test

TRAIN candidate ↔ truth; losses**INFER** prediction ↔ prediction; NMS**EVAL** prediction ↔ truth; AP

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almost no positive anchors	assignment geometry too strict	histogram best IoU per truth; inspect τ_+
objectness predicts background	easy negatives dominate	log pos/neg loss mass; compare focal or sampling

Inspect geometry and assignment before changing the backbone: a model cannot learn targets that the pipeline encoded incorrectly.

Follow along: boxes, IoU, matching, and anchor round-trips

I INTERACTIVE

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

INTERACTIVE

↗ colab.research.google.com/github/nipunbatra/dl-teaching/blob/master/notebooks/L09/01_iou_and_bbox_losses.ipynb

Open the exact Colab. Predict the $\frac{1444}{1756}$ IoU, implement the clamp, verify the anchor encode/decode round-trip, and build the held matching ledger before reading its outputs.

Follow along: boxes, IoU, matching, and anchor round-trips

I INTERACTIVE

TRAIN candidate $\leftrightarrow$ truth; losses

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Open the exact Colab. Predict the $\frac{1444}{1756}$ IoU, implement the clamp, verify the anchor encode/decode round-trip, and build the held matching ledger before reading its outputs.

BOX

$$[x_0, y_0, x_1, y_1]; \text{no} + 1$$

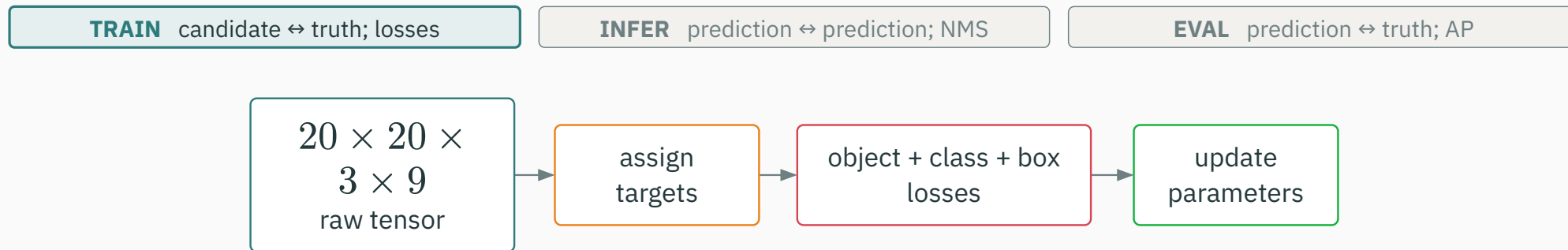
OVERLAP

$$\text{IoU}(A, B) \approx .822$$

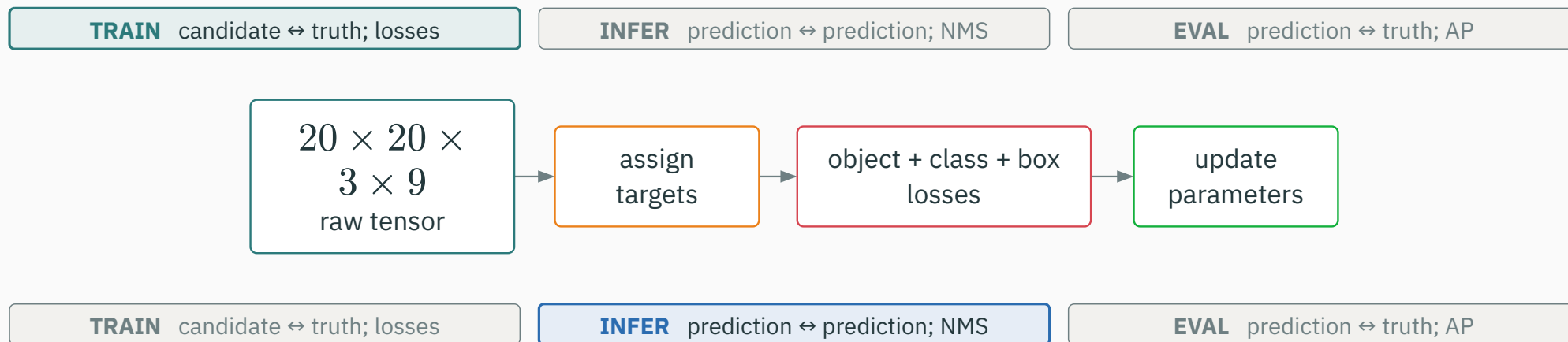
ANCHOR

$$t = (.20, -.10, .693, 0)$$

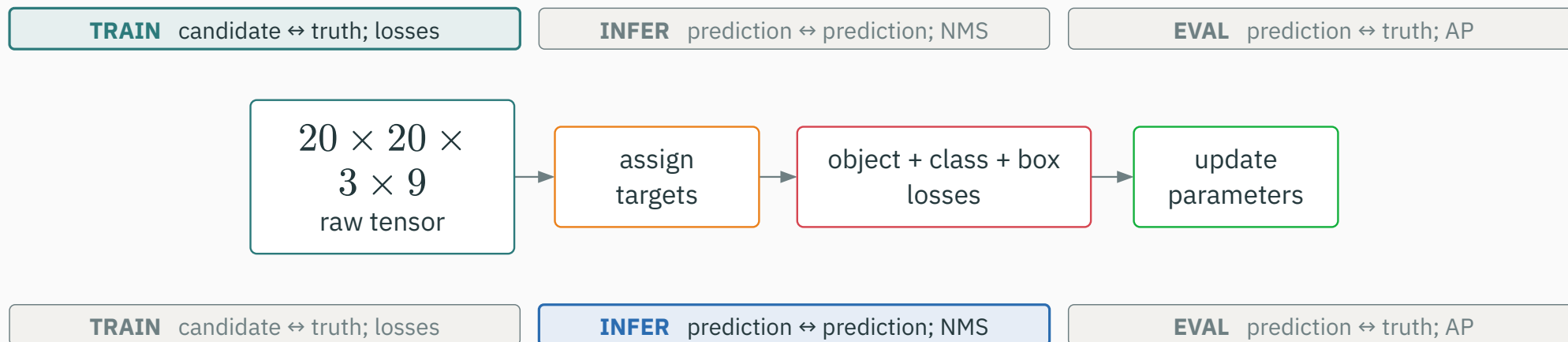
TRAIN closes with targets; inference opens without them



TRAIN closes with targets; inference opens without them



TRAIN closes with targets; inference opens without them



At inference there is no truth, no assignment, and no loss—only decoded boxes, scores, thresholds, and prediction-to-prediction suppression.

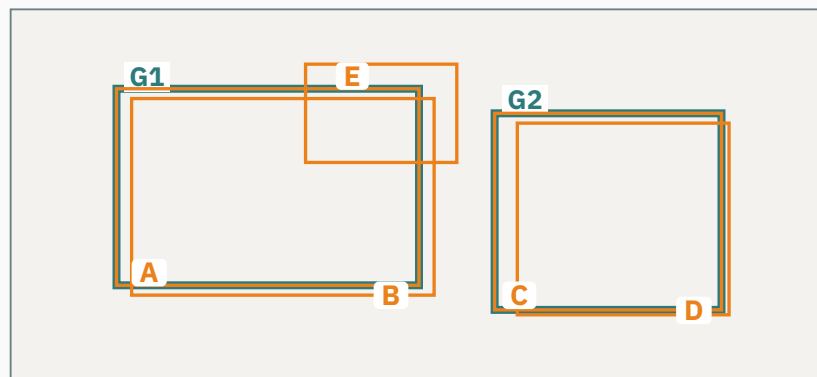
INFER: score first, then suppress duplicates

The raw head cannot know which high score is redundant

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



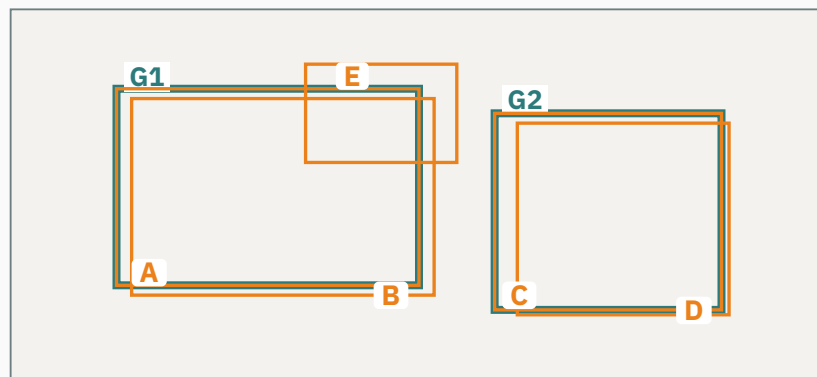
A .95 B .90 E .89 C .88 D .75

The raw head cannot know which high score is redundant

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



A .95 B .90 E .89 C .88 D .75

A and B describe cup 1; C and D describe cup 2; E is a confident isolated false alarm. Inference does not have those truth labels.

Score thresholding removes weak evidence—not duplicates

QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

Use a deliberately permissive score threshold $s \geq .70$.

A .95

B .90

E .89

C .88

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Score thresholding removes weak evidence—not duplicates

TRAIN candidate $\leftrightarrow$ truth; losses

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E .89

C .88

D .75

All five survive.

Score thresholding removes weak evidence—not duplicates

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TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

Use a deliberately permissive score threshold $s \geq .70$.

A .95

B .90

E .89

C .88

D .75

All five survive.

Thresholding can drop low scores, but it cannot tell that two high-scoring boxes describe the same object.

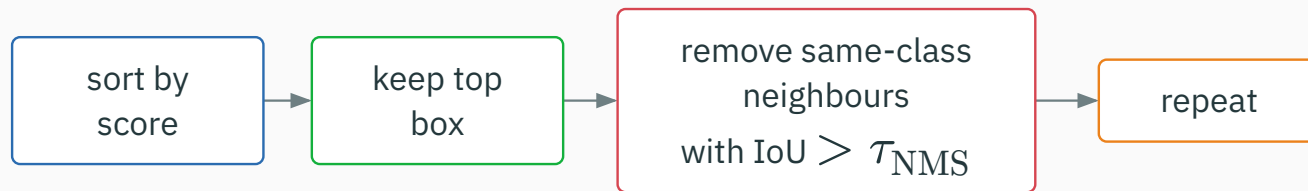
NMS is a greedy prediction-to-prediction comparison

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



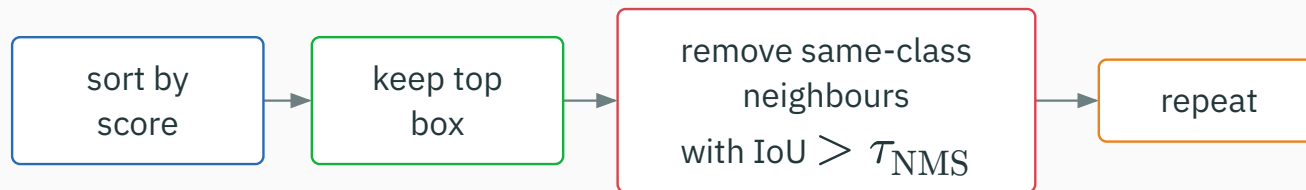
NMS is a greedy prediction-to-prediction comparison

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



No ground truth participates. Standard NMS has no learned parameter and no gradient; here $\tau_{\text{NMS}} = .50$.

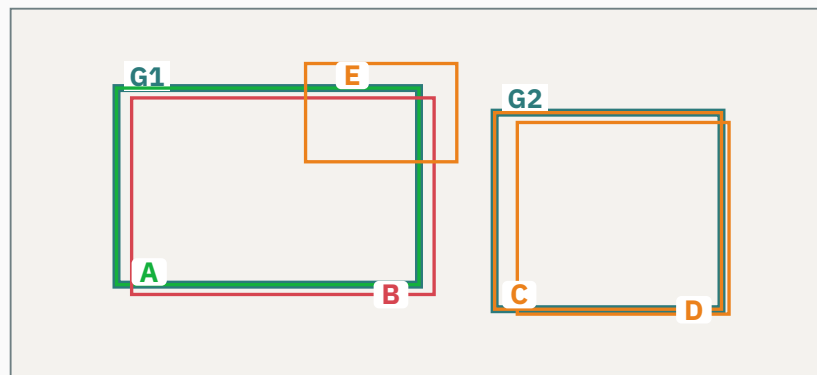
Step 1: A claims its high-overlap neighbourhood

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



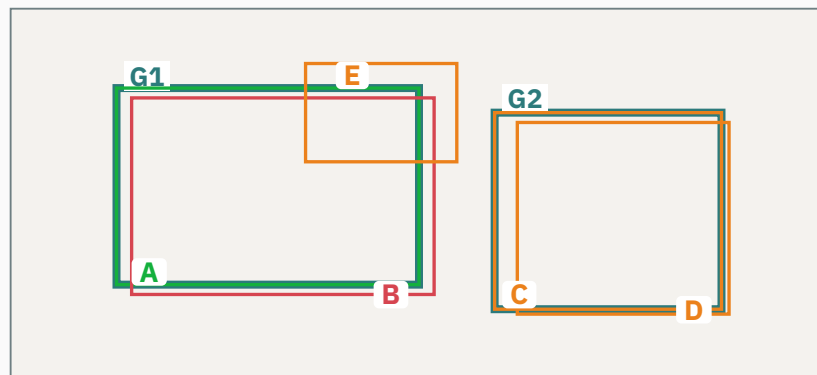
Step 1: A claims its high-overlap neighbourhood

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



$$\text{IoU}(A, B) = .822 > .50 \rightarrow \text{keep A, suppress B}$$

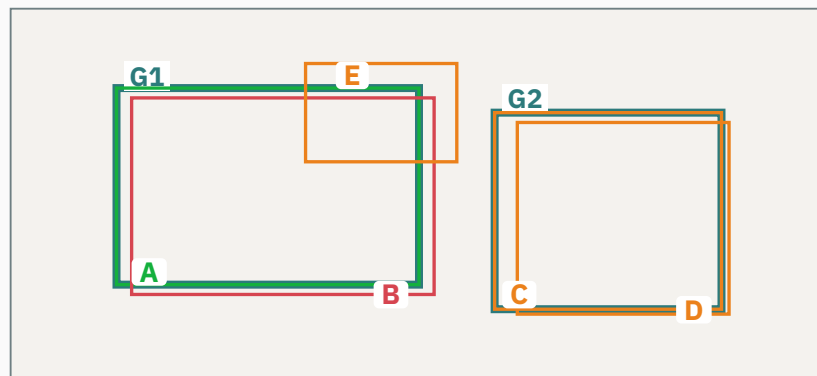
Step 1: A claims its high-overlap neighbourhood

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



$$\text{IoU}(A, B) = .822 > .50 \rightarrow \text{keep } A, \text{ suppress } B$$

E, C, and D remain because none overlaps A above the NMS threshold.

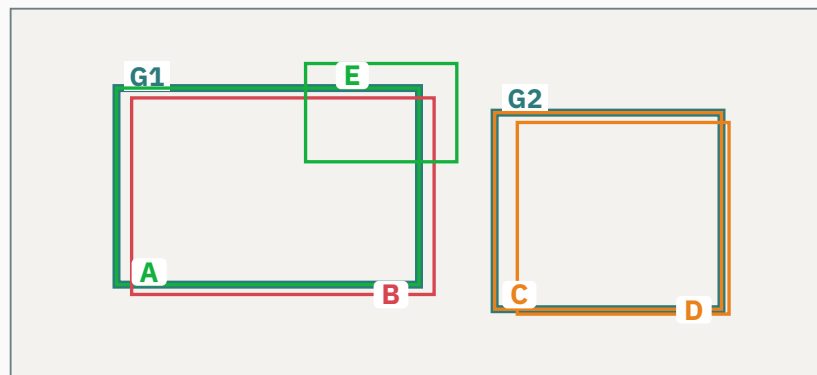
Step 2: E survives because NMS cannot test correctness

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



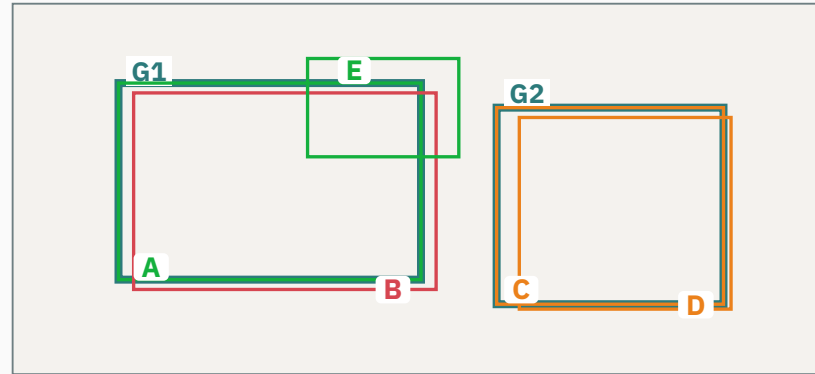
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TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



$E = .89$ is now the highest remaining score; its IoU with every survivor is below .50.

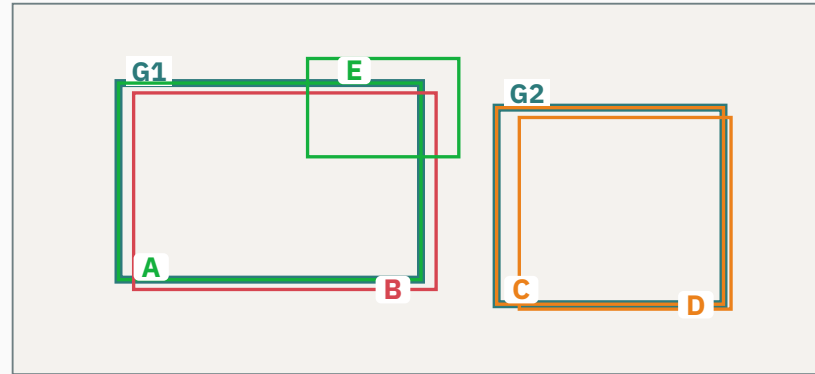
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EVAL prediction $\leftrightarrow$ truth; AP



$E = .89$ is now the highest remaining score; its IoU with every survivor is below .50.

Keep E. NMS removes duplicates around a stronger prediction; it does not remove isolated false alarms.

Step 3: C claims the second cup neighbourhood

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

HELD CASE · LAST PAIR $C = [60, 15, 90, 55]$ and $D = [63, 17, 91, 56]$.

$$A_I = (90 - 63)(55 - 17) = 27 \cdot 38 = 1,026$$

Step 3: C claims the second cup neighbourhood

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$$\text{IoU}(C, D) = \frac{1026}{1266} \approx .810 > .50$$

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$$\text{IoU}(C, D) = \frac{1026}{1266} \approx .810 > .50$$

Keep C; suppress D. No candidates remain.

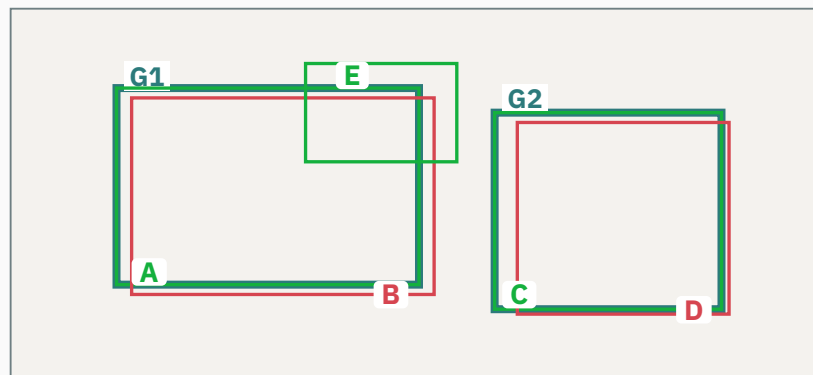
NMS returns A, E, C—not the two truths

A ANSWER

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



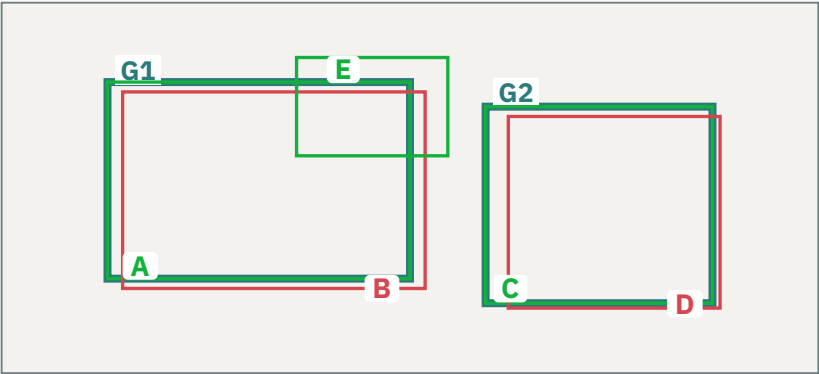
NMS returns A, E, C—not the two truths

A ANSWER

TRAIN candidate ↔ truth; losses

INFER prediction ↔ prediction; NMS

EVAL prediction ↔ truth; AP



ID	score	box $[x_0, y_0, x_1, y_1]$	status now	next comparison
A	.95	$[10, 10, 50, 50]$	kept	suppressed B
E	.89	$[35, 5, 55, 25]$	kept	isolated
C	.88	$[60, 15, 90, 55]$	kept	suppressed D

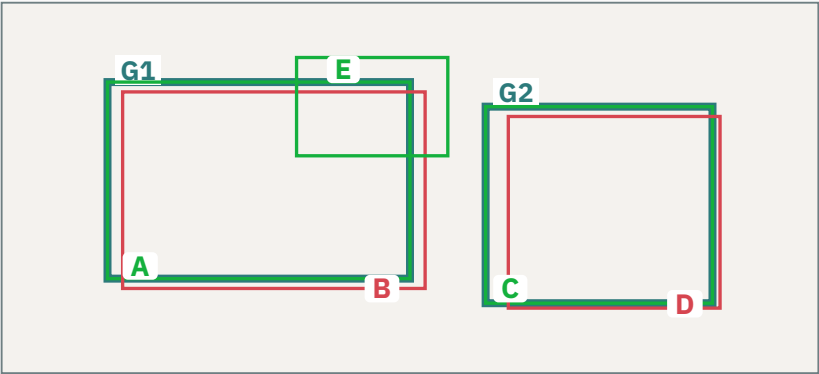
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Inference produced a **non-redundant** set. Whether every member is correct is an evaluation question.

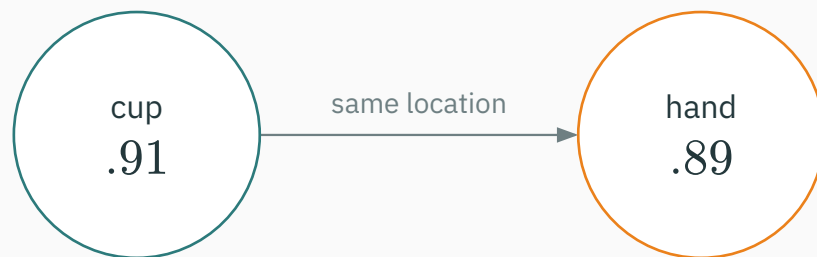
Class-aware NMS prevents a cup from deleting a hand

QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



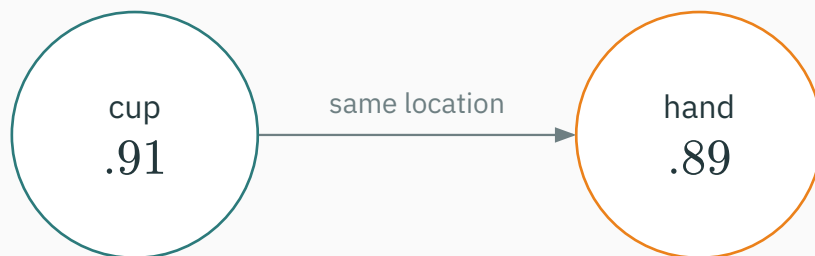
Class-aware NMS prevents a cup from deleting a hand

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EVAL prediction $\leftrightarrow$ truth; AP



Usually compare boxes within the same class. Cross-class overlap may describe two real, interacting objects.

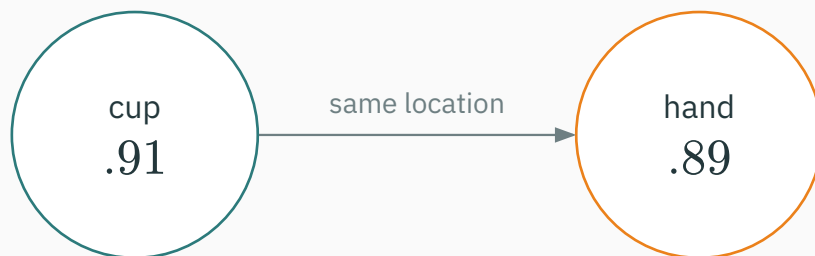
Class-aware NMS prevents a cup from deleting a hand

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EVAL prediction $\leftrightarrow$ truth; AP



Usually compare boxes within the same class. Cross-class overlap may describe two real, interacting objects.

A low NMS threshold can still merge two nearby objects of the **same** class. Increasing τ_{NMS} often keeps more boxes for a fixed score order, but keep-count is not a global monotonic law across changed scores/classes/pipelines.

Follow along: print every NMS decision, then break it

I INTERACTIVE

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

INTERACTIVE

↗ colab.research.google.com/github/nipunbatra/dl-teaching/blob/master/notebooks/L09/02_nms_from_scratch.ipynb

Open the exact Colab. Predict A, E, C; print the keep/suppress trace; verify class-aware NMS; then construct two distinct same-class objects that a low threshold wrongly merges.

Follow along: print every NMS decision, then break it

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Open the exact Colab. Predict A, E, C; print the keep/suppress trace; verify class-aware NMS; then construct two distinct same-class objects that a low threshold wrongly merges.

KEEP

A .95, E .89, C .88

SUPPRESS

B by A; D by C

LIMIT

isolated false alarm E survives

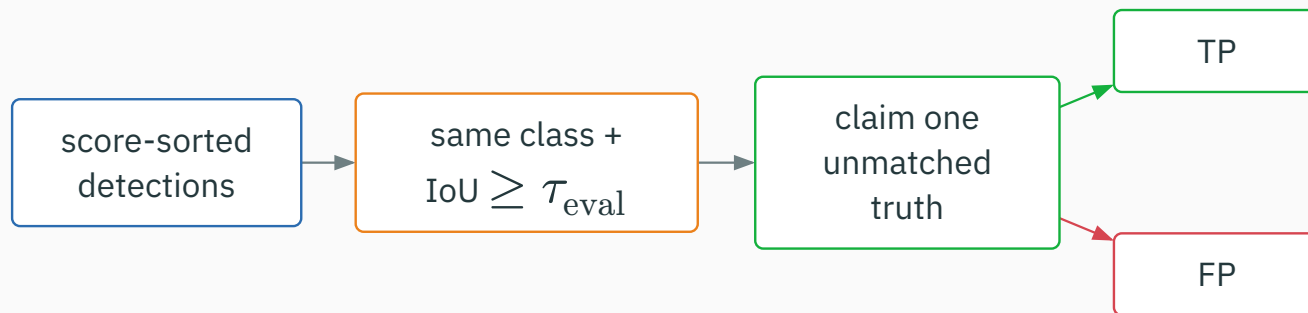
EVAL: match the returned set to truth, one claim at a time

Evaluation matching is not training assignment

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

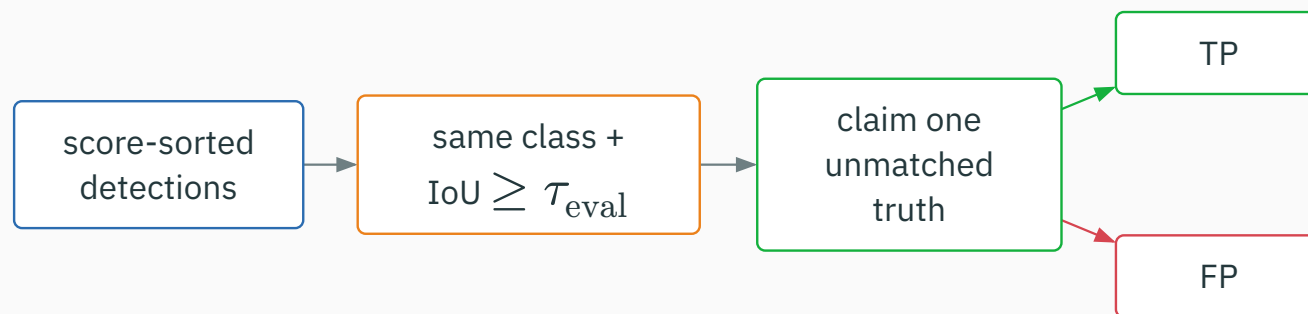


Evaluation matching is not training assignment

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



At $\tau_{\text{eval}} = .50$, each truth may be claimed once. A duplicate would be an FP even if its IoU were high.

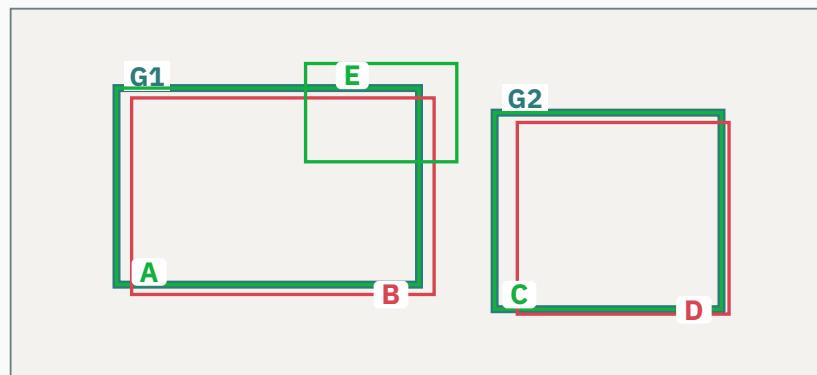
Commit: which of A, E, C are true positives?

QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



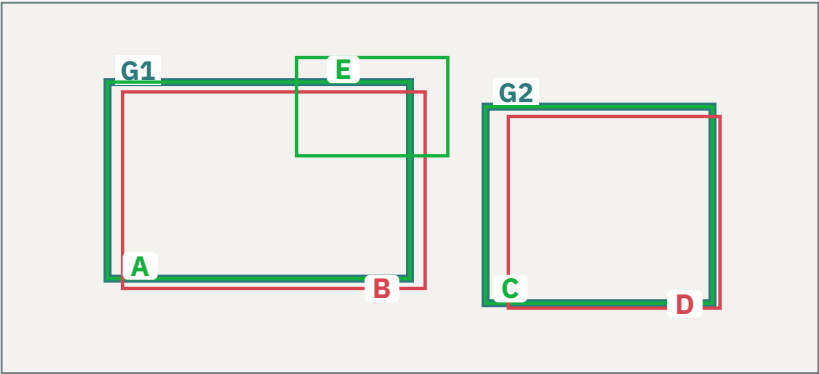
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QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



detection	score	best same-class IoU	commit
A	.95	1.000 with G_1	?
E	.89	.127 with G_1	?
C	.88	1.000 with G_2	?

Commit: which of A, E, C are true positives?

TRAIN

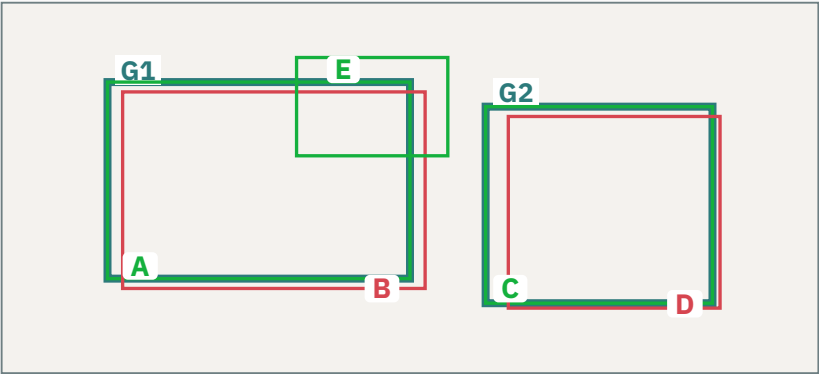
candidate ↔ truth; losses

INFER

prediction ↔ prediction; NMS

EVAL

prediction ↔ truth; AP



detection	score	best same-class IoU	commit
A	.95	1.000 with G_1	?
E	.89	.127 with G_1	?
C	.88	1.000 with G_2	?

Process the score order. A truth claimed by an earlier detection is unavailable to later detections.

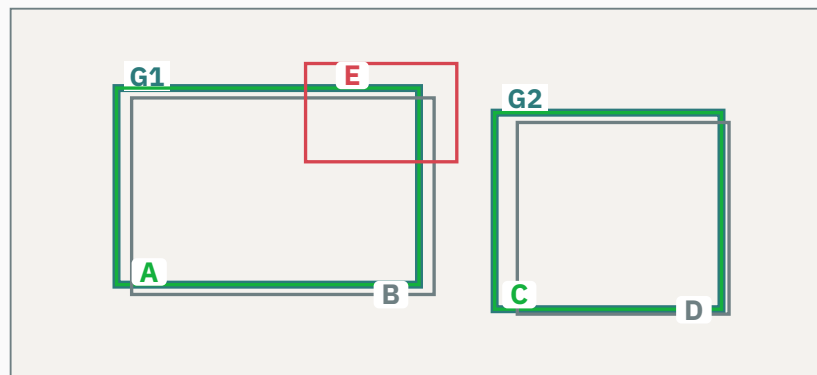
Evaluation exposes the false alarm NMS could not see

A ANSWER

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



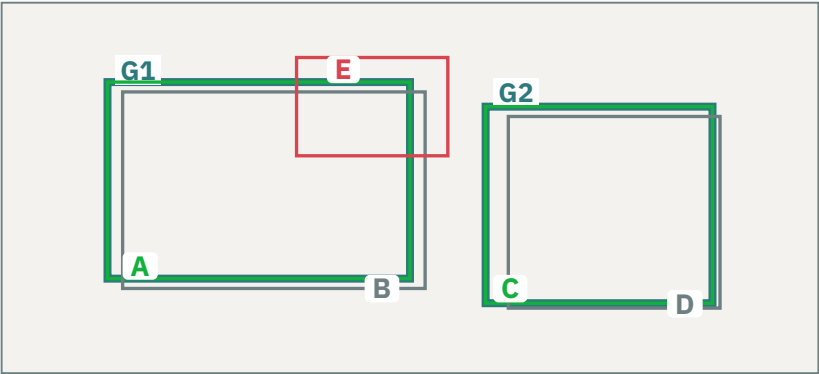
Evaluation exposes the false alarm NMS could not see

A ANSWER

TRAIN candidate ↔ truth; losses

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detection	claim	reason	result
A	G_1	$\text{IoU} = 1.000$	TP
E	none	$\text{IoU} = .127 < .50$	FP
C	G_2	$\text{IoU} = 1.000$	TP

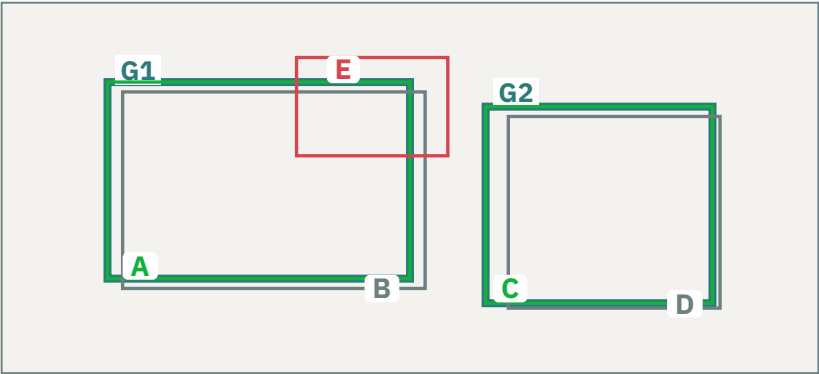
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The final set has two TPs, one FP, and zero FNs. “Non-redundant” did not mean “all correct.”

Precision and recall evolve with the score ranking

D DERIVATION

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EVAL prediction $\leftrightarrow$ truth; AP

after	cum. TP	cum. FP	precision	recall
A	1	0	$\frac{1}{1} = 1.000$	$\frac{1}{2} = .500$

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C	2	1	$\frac{2}{3} = .667$	$\frac{2}{2} = 1.000$

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Lowering the score threshold admitted E before C: precision fell, then recall rose when C arrived.

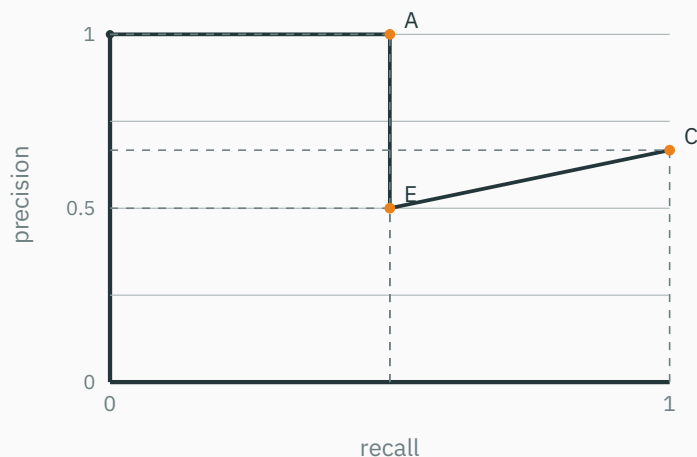
The toy AP convention averages precision at TP ranks

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At the two TP ranks:

$$p_A = 1$$

$$p_C = \frac{2}{3}$$

$$\text{AP}_{\text{toy}} = \frac{1 + \frac{2}{3}}{2} = \frac{5}{6} \approx .833$$

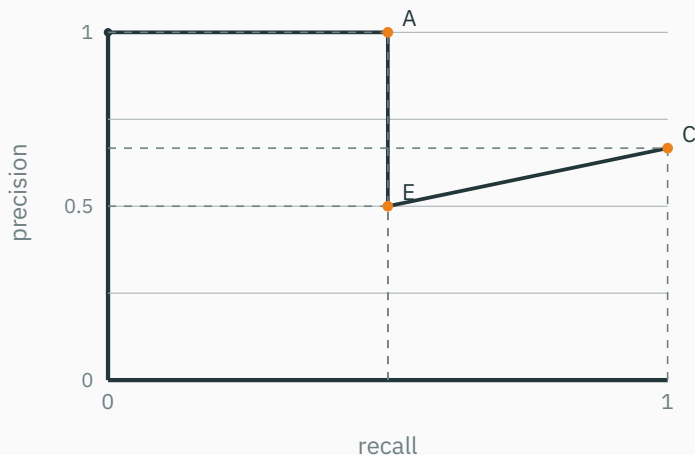
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After NMS: A, E, C; B and D were suppressed before evaluation. Axes run from 0 to 1.

This TP-rank average is a transparent teaching convention. It is not a claim about every benchmark's interpolation rule.

SHOULD · COCO AP fixes the full evaluation protocol

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

LOCALIZATION

10 IoU thresholds .50, .55, ..., .95

INTERPOLATION

101 recall thresholds 0, .01, ..., 1

AGGREGATION

average over classes; declare area +
max detections

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Say AP50 for one IoU cutoff. “COCO AP” normally means the average across IoU thresholds; the official API also declares area ranges and maxDets.

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Official COCO evaluation implementation

Diagnose metric surprises on the correct clock

TRAIN candidate $\leftrightarrow$ truth; losses**INFER** prediction $\leftrightarrow$ prediction; NMS**EVAL** prediction $\leftrightarrow$ truth; AP

symptom	suspect	smallest discriminating test
high scores; low AP	ranking is confidently wrong	inspect first FP for class, IoU, duplicate status
AP50 high; COCO AP low	boxes are loose	compare AP50 vs AP75; histogram matched IoU
recall caps below 1	thresholding / missing candidates	lower score threshold before NMS; count unmatched truths

TRAIN candidate ↔ truth; losses

INFER prediction ↔ prediction; NMS

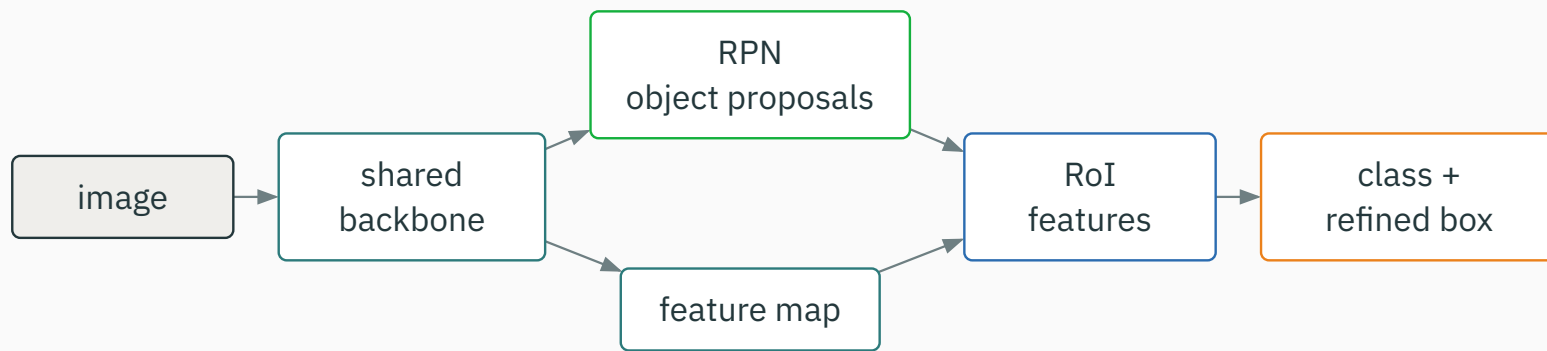
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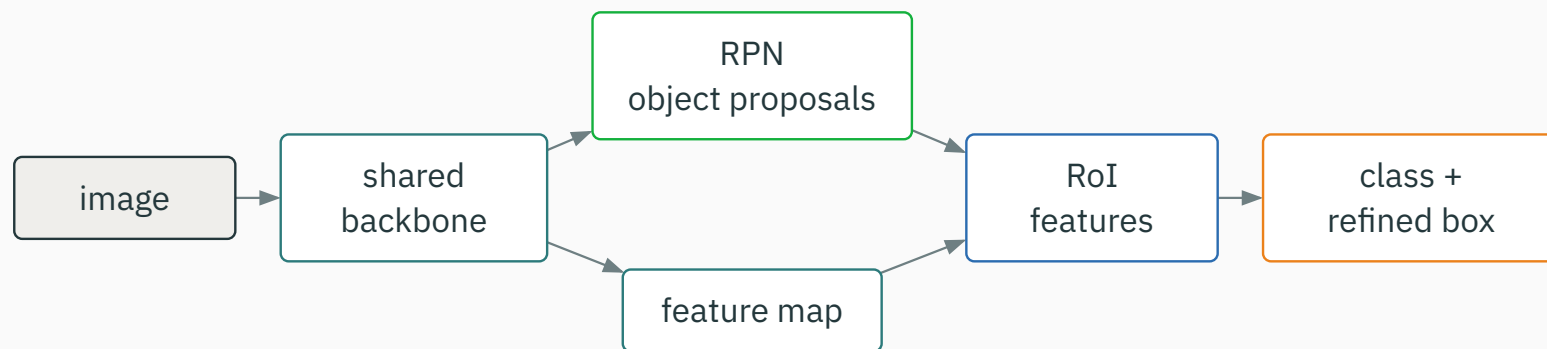
Loss, NMS, and AP expose different failure surfaces. Name the clock before changing a threshold.

SHOULD: architectures rearrange the same evidence

SHOULD · two-stage means propose, then refine



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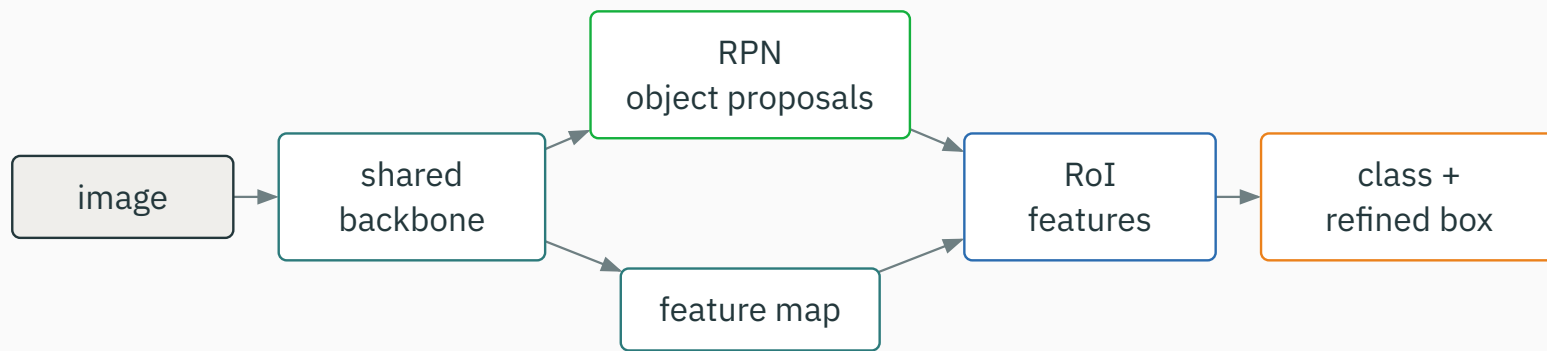
RPN

Region Proposal Network: predicts objectness + proposal boxes densely.

ROI

Region of Interest: samples a fixed-size feature for each proposal.

SHOULD · two-stage means propose, then refine



RPN

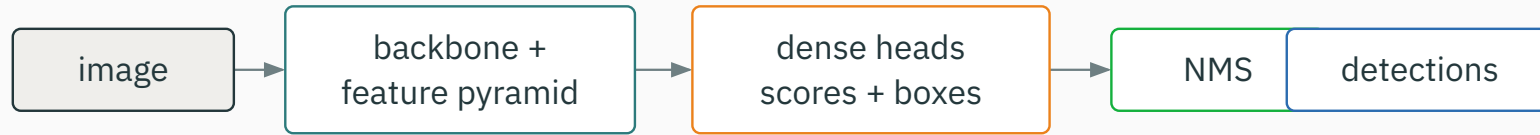
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Region Proposal Network: predicts objectness + proposal boxes densely.

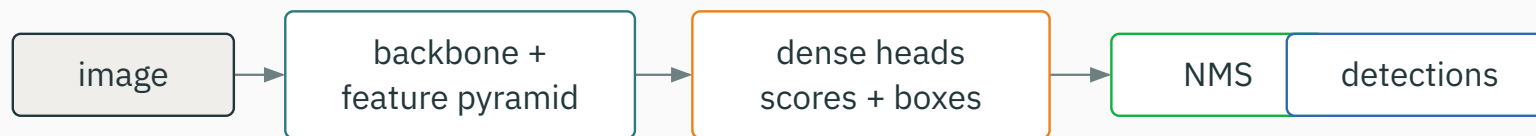
Region of Interest: samples a fixed-size feature for each proposal.

Stage 1 asks “where might an object be?” Stage 2 spends more computation to classify and refine a smaller proposal set.

SHOULD · one-stage predicts every candidate directly

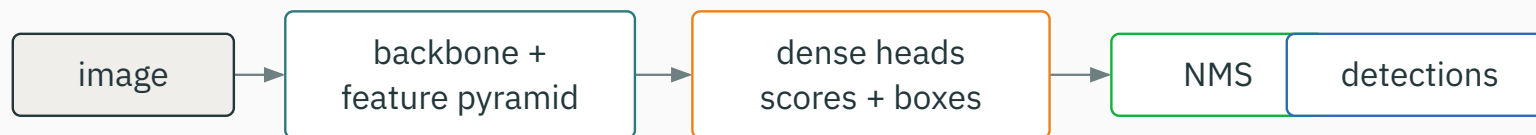


SHOULD · one-stage predicts every candidate directly



There is no learned proposal stage or per-RoI second head: classification and localization happen densely in one pass.

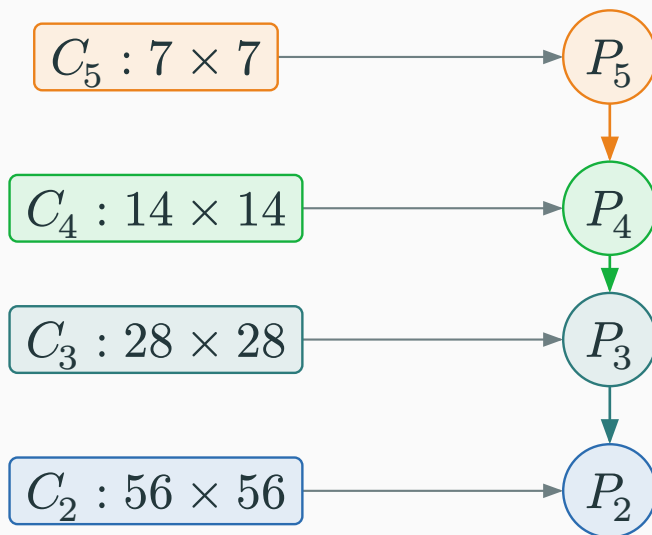
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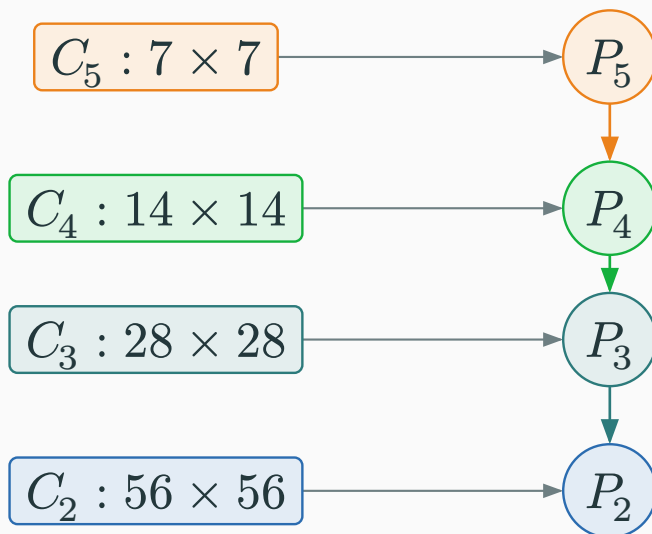
There is no learned proposal stage or per-RoI second head: classification and localization happen densely in one pass.

One-stage versus two-stage describes where proposal/refinement computation occurs—not a timeless guarantee about speed or accuracy.

SHOULD · FPN gives small and large objects strong features

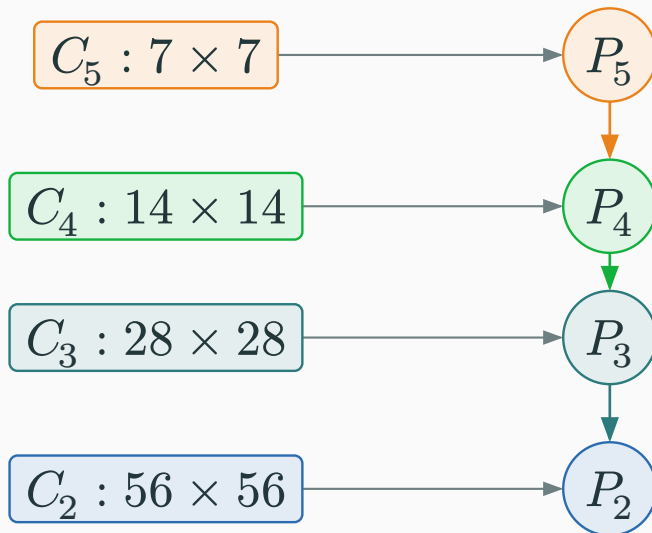


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FPN = **Feature Pyramid Network**: lateral 1×1 projections + top-down upsampling + same-shape addition.

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FPN = **Feature Pyramid Network**: lateral 1×1 projections + top-down upsampling + same-shape addition.

High-resolution levels localize small objects; top-down semantics make those levels useful for recognition.

ANCHOR-BASED

start from (x_a, y_a, w_a, h_a) ; regress normalized translations and log scales

ANCHOR-FREE

predict a center/keypoint or distances to left, top, right, bottom sides

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Both still need spatial features, class evidence, box geometry, a training matcher, and an evaluation protocol.

SOFT-NMS

decay a neighbour's score instead of deleting it; nearby objects may survive

SET PREDICTION

train a fixed query set with one-to-one assignment;
duplicate avoidance moves into learning

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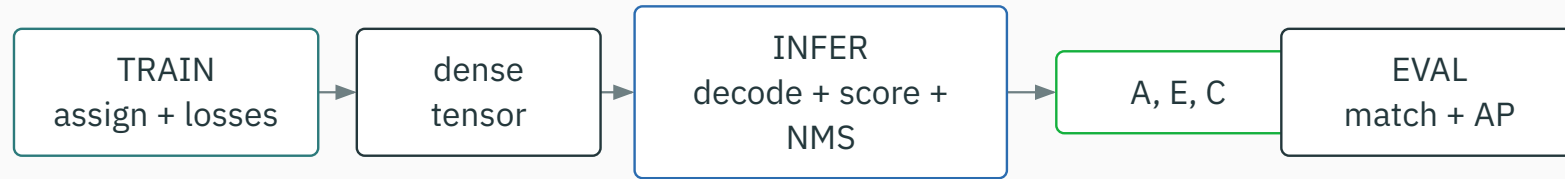
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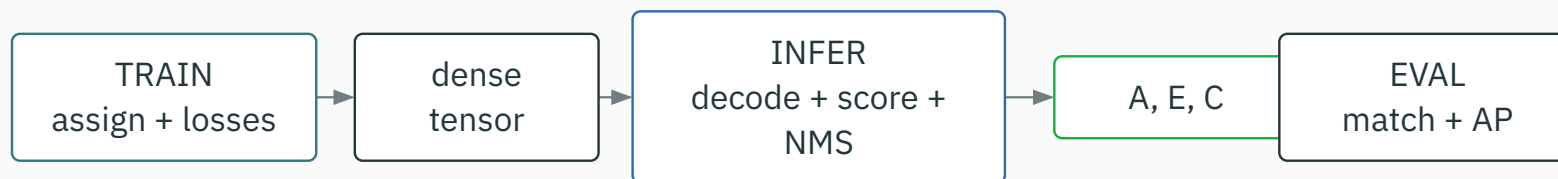
These alternatives change inference and training, but the evaluation clock still matches scored predictions to truths under a declared protocol.

One pipeline, three clocks, one returned set

The complete pipeline changes what IoU is comparing

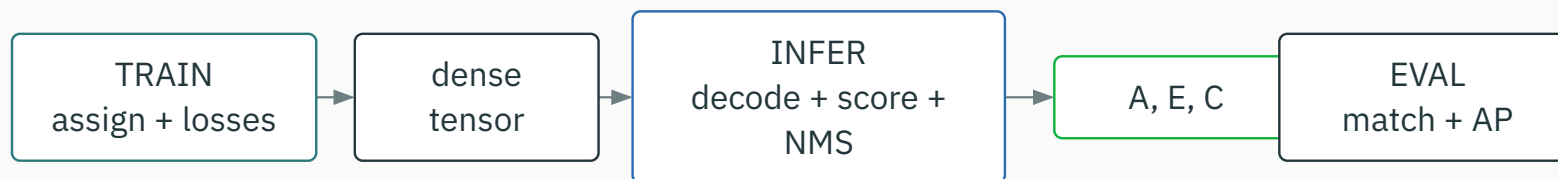


The complete pipeline changes what IoU is comparing



clock	IoU compares	decision
TRAIN	candidate ↔ truth	which targets and losses apply
INFER	prediction ↔ prediction	which neighbours are redundant
EVAL	prediction ↔ truth	TP / FP / FN and AP

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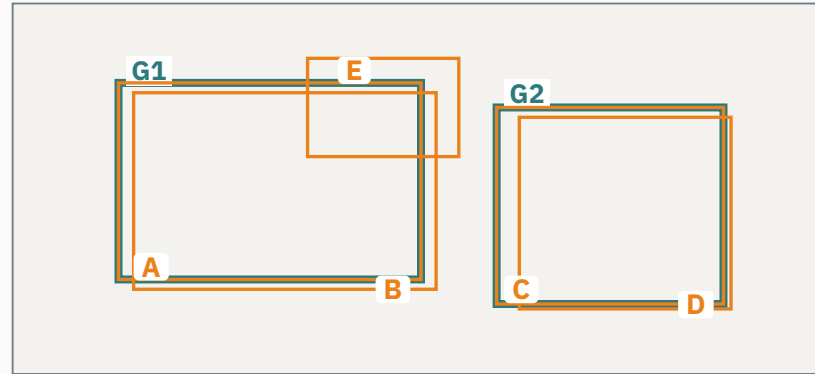
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The formula is the same; the objects, threshold, and consequence are not.

Revisit the opening commitment before revealing it

QUESTION

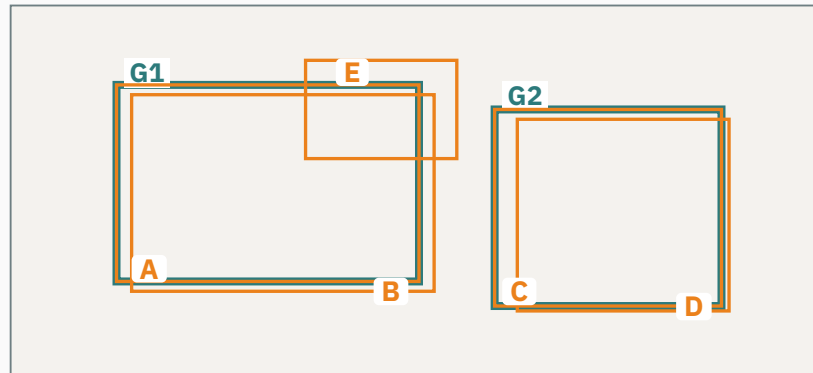
HELD CASE · SAME FIVE CANDIDATES A .95, B .90, E .89, C .88, D .75; $\tau_{\text{NMS}} = .50$, $\tau_{\text{eval}} = .50$.



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A · THRESHOLD

all five pass .70

B · GEOMETRY

decode → score → NMS → evaluation
match

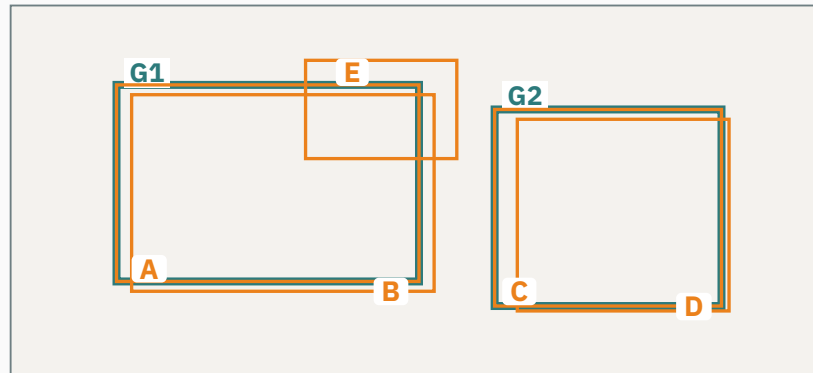
C · LOSS

requires truths unavailable at
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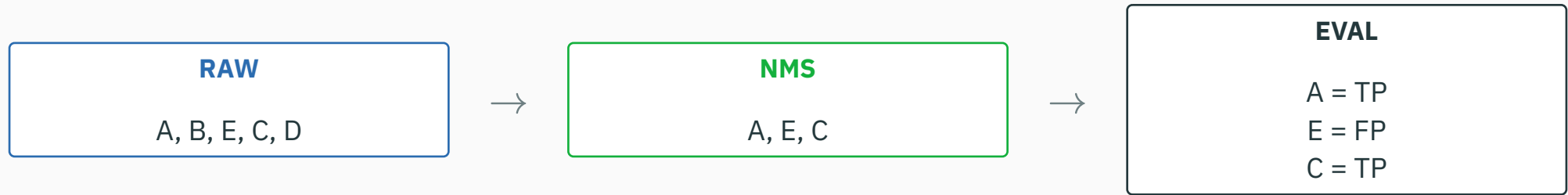
C · LOSS

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Defend your original choice using the exact boxes and clocks—not a detector name.

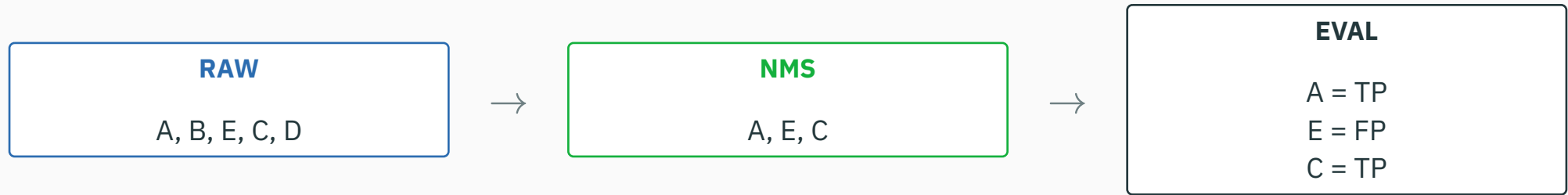
Answer: B is complete—and still cannot remove E

A ANSWER



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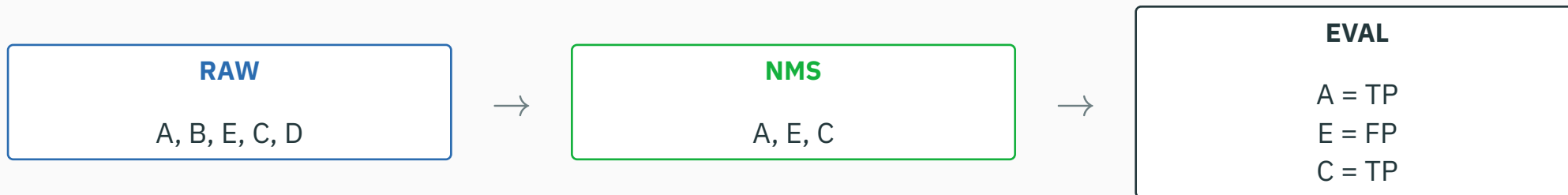
A ANSWER



$$AP_{\text{toy}} = \frac{5}{6} \approx .833$$

Answer: B is complete—and still cannot remove E

A ANSWER



$$AP_{\text{toy}} = \frac{5}{6} \approx .833$$

Thresholds control admission, NMS controls redundancy, and matching measures correctness. No one operation can substitute for the others.

Given a new detector, ask in order:

1. What box coordinate and area convention does every component use?
2. What is the dense output shape, and what does each scalar mean?
3. Which TRAIN matcher creates positive, ignored, and background targets?
4. Which loss reduction and weights determine gradient mass?
5. Which INFER score and NMS policies create the returned set?
6. Which EVAL IoU, interpolation, class, area, and max-detection protocol defines AP?
7. Which symptom → suspect → test check will distinguish geometry from learning failure?

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A detector is trustworthy only when the tensor, geometry, three clocks, and metric protocol are all explicit.

L11 handoff: boxes become a prediction at every pixel

DETECTION

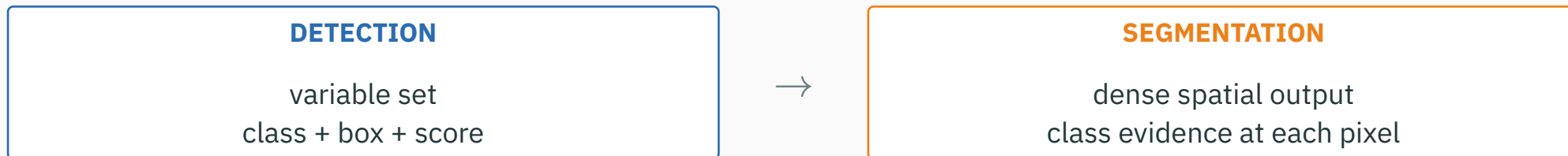
variable set
class + box + score



SEGMENTATION

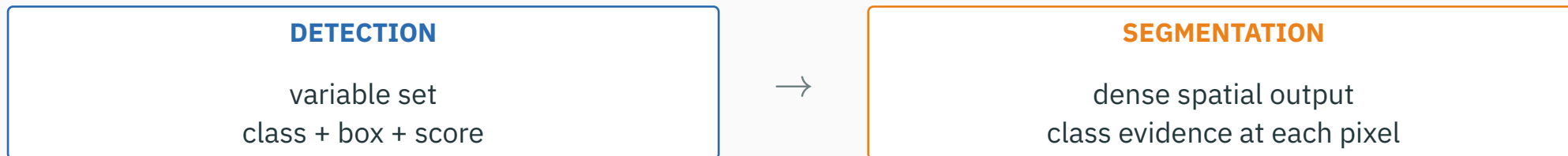
dense spatial output
class evidence at each pixel

L11 handoff: boxes become a prediction at every pixel



Both reuse the L8B backbone. L11 will replace four-number rectangles with masks, then recover boundaries that pooling made coarse.

L11 handoff: boxes become a prediction at every pixel



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Today asked “which objects, and where?” Next asks “which class owns every pixel—and which instance?”

DETECTOR FAMILIES

Girshick et al. · R-CNN
Girshick · Fast R-CNN
Ren et al. · Faster R-CNN
Redmon et al. · YOLO

LOSSES, PYRAMIDS + CODE

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Use the papers for historical claims; use the held five-box ledger and exact notebooks for reproducible arithmetic.